

Question 1

R

```
library(ggplot2)
```

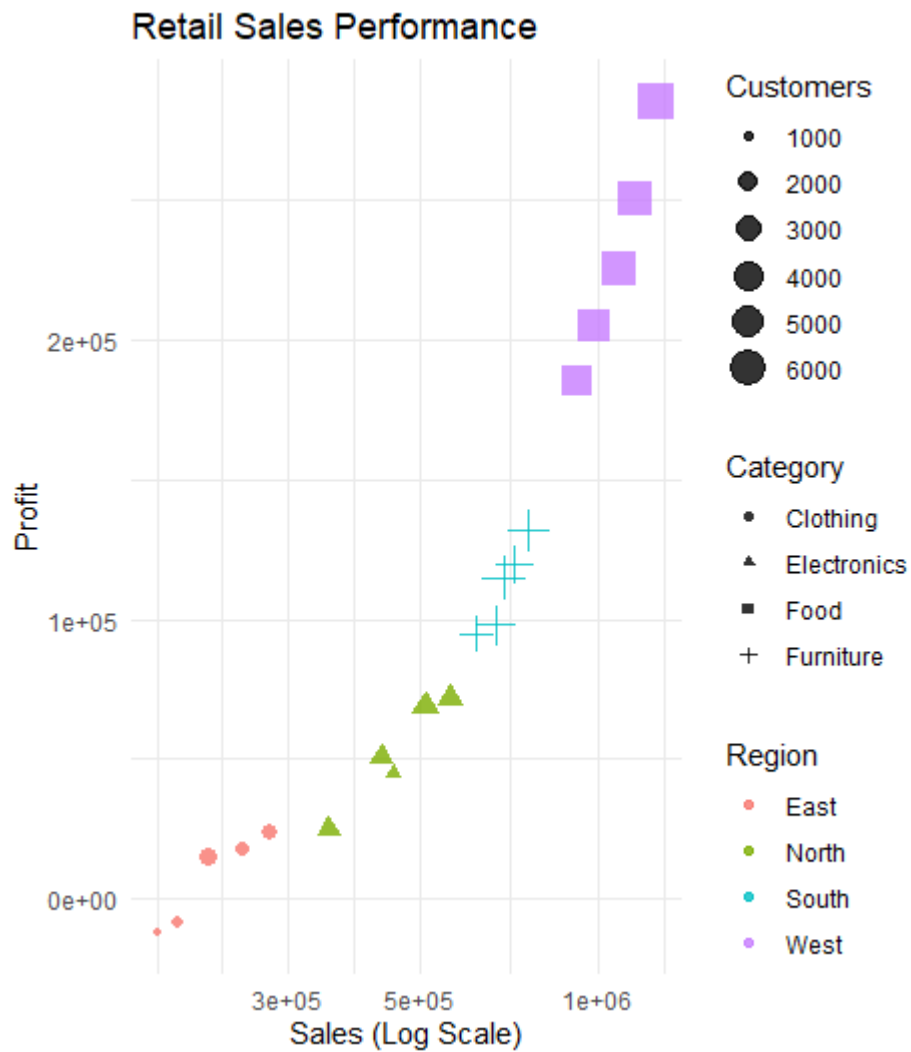
```
retail <- data.frame(  
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",  
  "B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),  
  Region=c("North","South","East","West","North","South","East","West",  
  "North","South","East","West","North","South","East","West",  
  "North","South","East","West"),  
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",  
  "Furniture","Clothing","Food","Electronics","Furniture",  
  "Clothing","Food","Electronics","Furniture","Clothing",  
  "Food","Electronics","Furniture","Clothing","Food"),  
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,  
  560000,670000,195000,980000,430000,760000,280000,1150000,  
  510000,690000,220000,1250000),  
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,  
  72000,98000,-8000,205000,51000,132000,24000,250000,  
  69000,115000,15000,285000),  
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,  
  2100,3300,1100,4700,1750,3600,1450,5600,  
  2200,3900,1600,6100)  
)
```

```
ggplot(retail,  
  aes(x=Sales,  
  y=Profit,
```

```
    color=Region,  
    shape=Category,  
    size=Customers))+  
geom_point(alpha=0.8)+  
scale_x_log10()+  
labs(title="Retail Sales Performance",  
      x="Sales (Log Scale)",  
      y="Profit",  
      color="Region",  
      shape="Category",  
      size="Customers")+  
theme_minimal()
```

Interpretation

- X-axis = Sales
- Y-axis = Profit
- Colour = Region
- Shape = Product Category
- Bubble Size = Customers
- Log transformation on Sales reduces skewness and improves comparison among branches.



Question 2

Cartesian Coordinate Plot

R

```
library(ggplot2)
```

```
library(tidyr)
```

```
energy <- data.frame(
```

```
  Month=month.abb,
```

```
  Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
```

```
  Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
```

```
Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),  
Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),  
Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)  
)
```

```
energy_long <- pivot_longer(  
  energy,  
  cols=-Month,  
  names_to="Source",  
  values_to="Generation"  
)
```

```
ggplot(energy_long,  
  aes(Month,Generation,color=Source,group=Source))+  
  geom_line(size=1)+  
  geom_point()+  
  theme_minimal()+  
  labs(title="Monthly Electricity Generation")
```

Polar Coordinate Plot

R

```
ggplot(energy_long,  
  aes(x=Month,y=Generation,fill=Source))+  
  geom_bar(stat="identity")+  
  coord_polar()+  
  theme_minimal()+  
  labs(title="Electricity Generation (Polar Coordinate)")
```

Interpretation

Cartesian Coordinate

- Best for identifying trends.
- Easy comparison between months.
- Accurate value estimation.

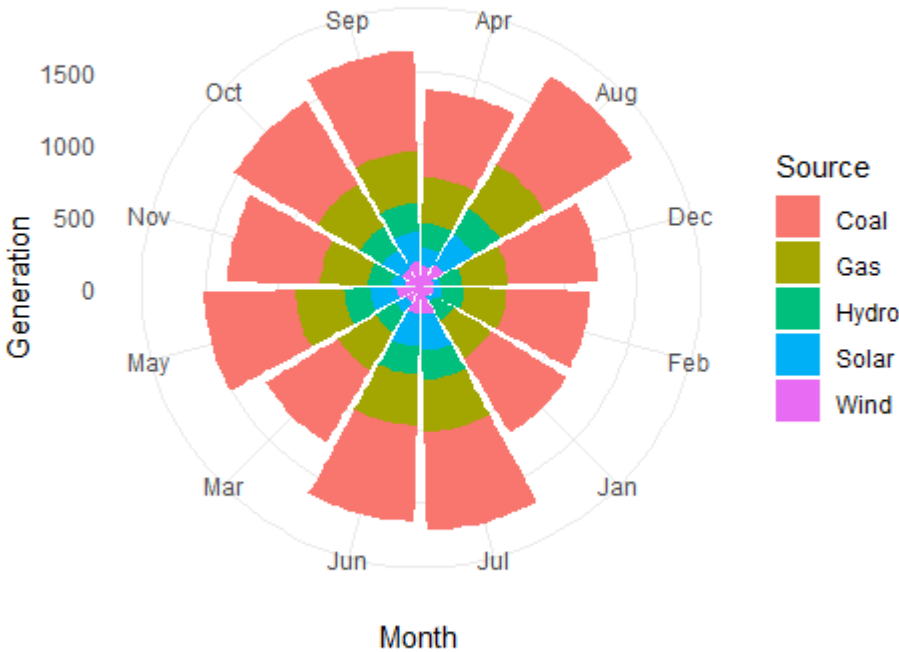
Polar Coordinate

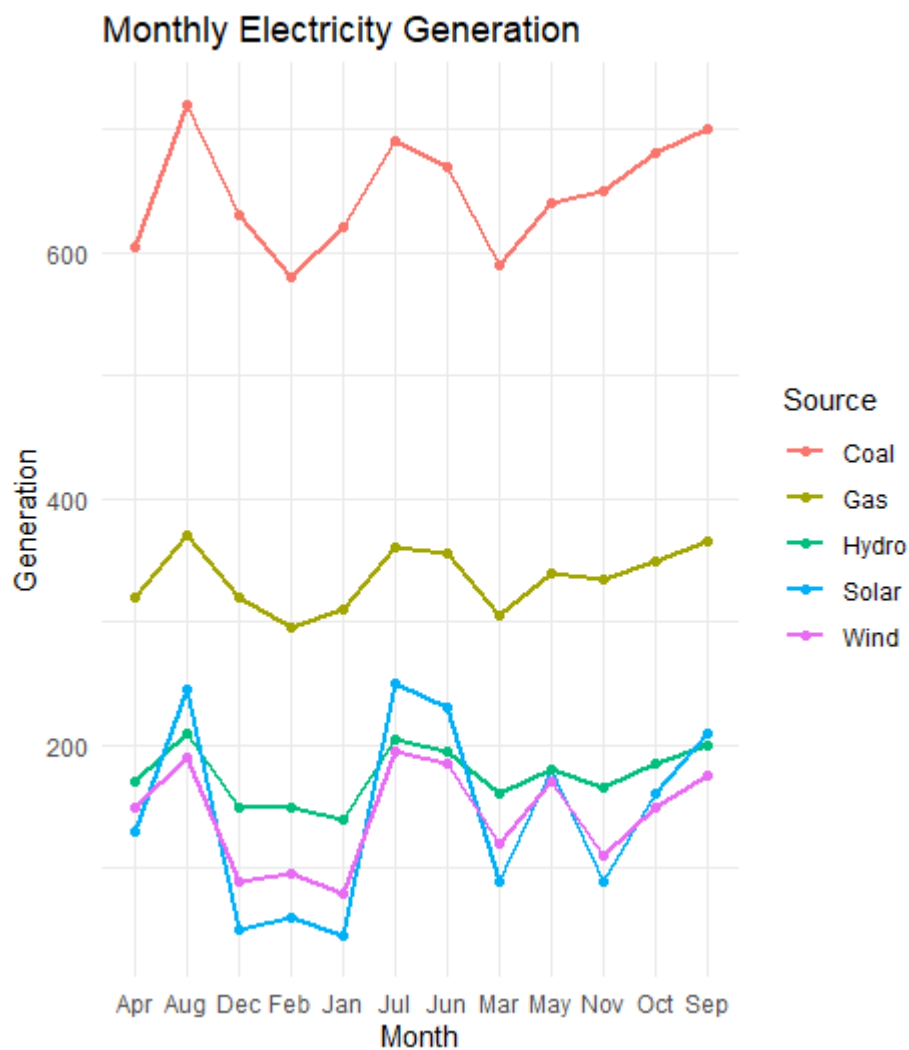
- Attractive visual.
- Good for cyclic data.
- Harder to compare exact values.

Conclusion

Cartesian coordinates provide better analytical insight because comparisons are more accurate.

Electricity Generation (Polar Coordinate)





Question 3

R

```
library(ggplot2)
```

```
climate <- data.frame(
```

```
City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",
```

```
"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
```

```
"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
```

```
"Surat","Visakhapatnam","Coimbatore","Mysuru"),
```

```

Temperature=c(44,35,38,40,39,31,33,43,45,39,
38,40,32,30,41,37,36,34,29,28),
Humidity=c(22,84,78,72,48,66,58,26,18,42,
36,45,89,91,30,34,76,82,68,64),
Rainfall=c(6,280,220,190,85,120,95,2,1,35,
48,72,360,410,12,20,180,240,130,145),
AQI=c(365,110,125,180,160,72,84,290,320,240,
175,220,58,52,270,190,145,98,60,55)
)

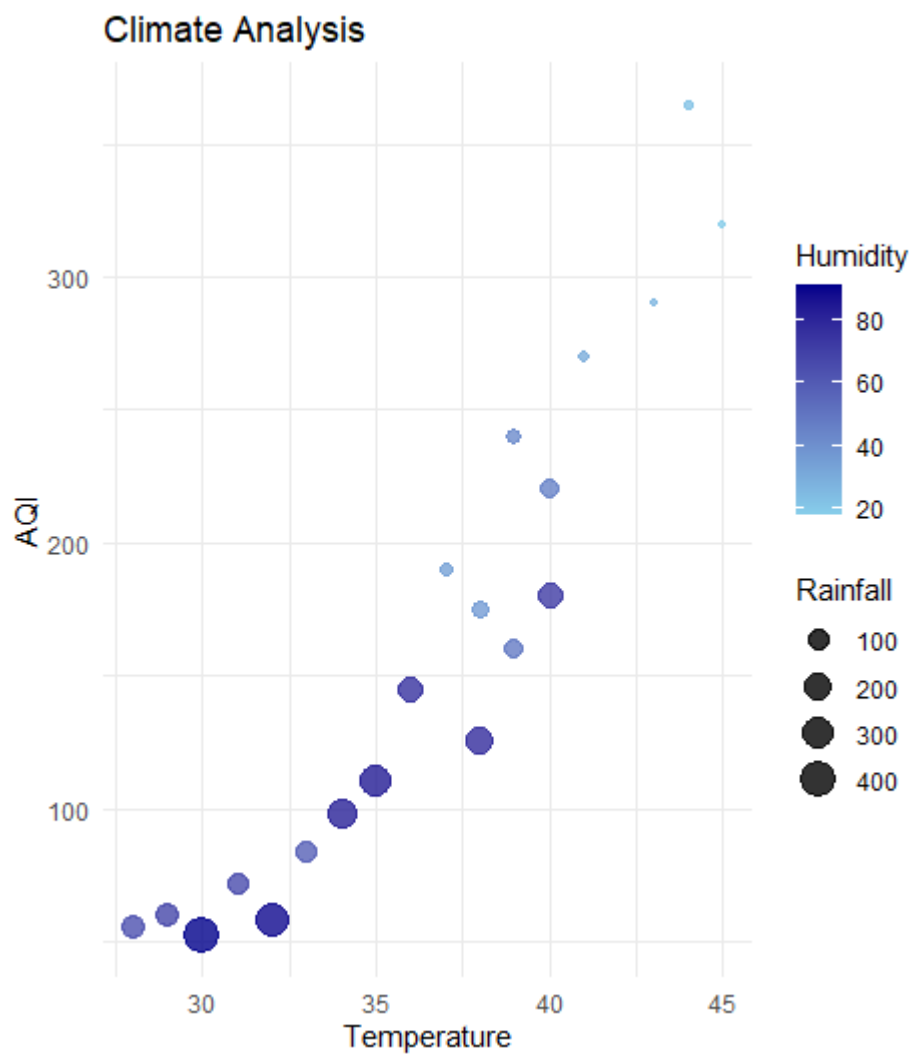
ggplot(climate,
aes(Temperature,
AQI,
size=Rainfall,
color=Humidity))+
geom_point(alpha=0.8)+
scale_color_gradient(low="skyblue",high="darkblue")+
labs(title="Climate Analysis",
x="Temperature",
y="AQI",
color="Humidity",
size="Rainfall")+
theme_minimal()

```

Interpretation

- X-axis = Temperature
- Y-axis = AQI
- Colour = Humidity
- Bubble Size = Rainfall

- Blue gradient clearly represents increasing humidity and is colorblind-friendly.



Question 4

R

```
library(ggplot2)
```

```
software <- data.frame(
  Team=paste("T",1:20,sep=""),
  LinesOfCode=c(12000,18000,25000,35000,48000,
    62000,81000,96000,120000,150000,
```

```

185000,240000,310000,390000,480000,
620000,810000,1050000,1350000,1800000),
Bugs=c(35,52,60,72,88,
110,125,142,165,188,
205,240,280,320,365,
420,490,560,640,720),
Developers=c(5,7,9,10,12,
14,16,18,20,22,
25,28,30,34,38,
42,46,50,55,60),
Complexity=c(2.1,2.4,2.7,3.0,3.2,
3.5,3.8,4.1,4.5,4.8,
5.1,5.4,5.8,6.2,6.6,
7.1,7.6,8.0,8.5,9.0)
)

```

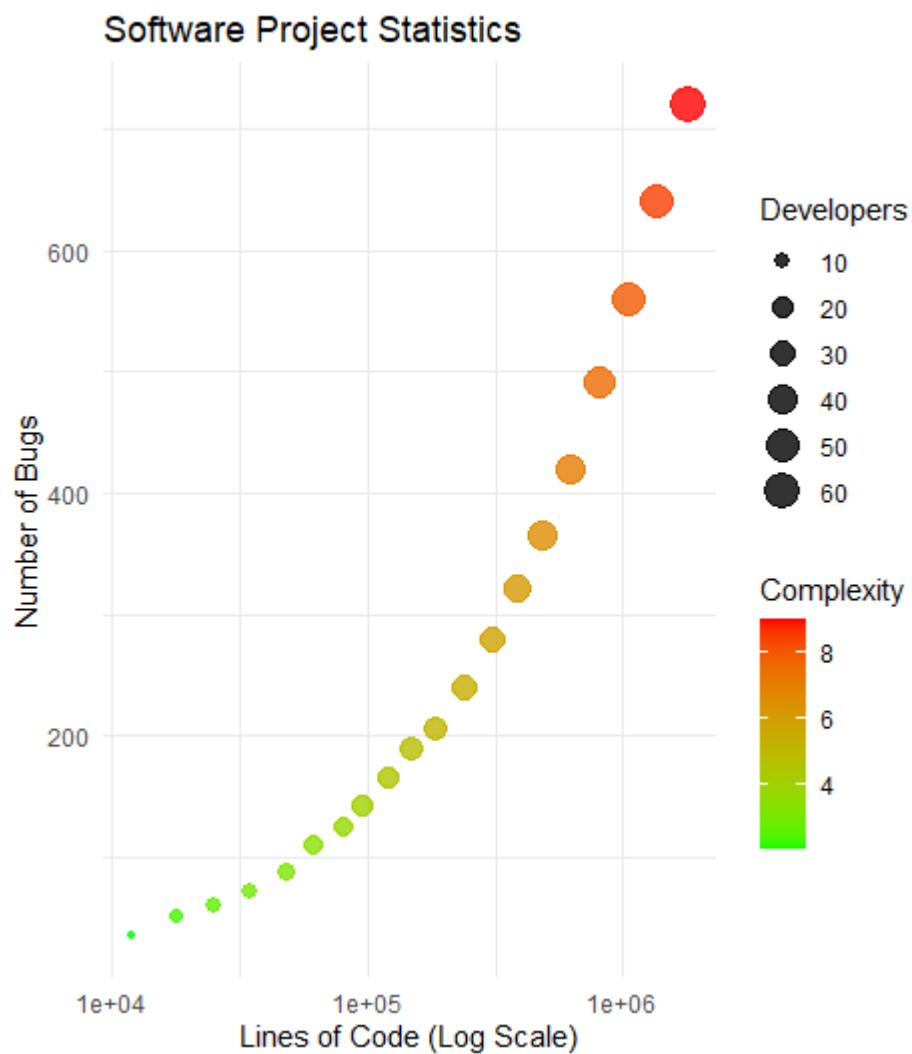
```

ggplot(software,
aes(LinesOfCode,
Bugs,
size=Developers,
color=Complexity))+
geom_point(alpha=0.8)+
scale_x_log10()+
scale_color_gradient(low="green",high="red")+
labs(title="Software Project Statistics",
x="Lines of Code (Log Scale)",
y="Number of Bugs")+
theme_minimal()

```

Interpretation

- Nonlinear transformation = Logarithmic Scale
- LOC ranges from 12,000 to 1,800,000, causing heavy skewness.
- Log scale spreads smaller values while compressing larger values.
- Makes comparison easier than a linear axis.



Question 5

R

```
library(ggplot2)
```

```
companies <- data.frame(
```

```

Company=paste("C",1:20,sep=""),
Sector=c("IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy"),
Revenue=c(420,580,720,850,980,
1150,1320,1480,1650,1820,
2050,2280,2550,2840,3150,
3480,3860,4250,4680,5200),
ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
20.4,17.8,14.2,10.5,22.1,
24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),
Employees=c(1500,3200,2100,4200,2800,
5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,
11800,13200,8600,14500,16000),
MarketCap=c(8500,12000,9800,7600,14500,
18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
42500,46800,35600,29800,52000)
)

```

```

ggplot(companies,
aes(Revenue,
ProfitMargin,
size=MarketCap,
color=Sector))+

```

```
geom_point(alpha=0.8)+  
scale_size_continuous(range=c(3,12))+  
labs(title="Company Financial Performance",  
x="Revenue",  
y="Profit Margin (%)",  
size="Market Cap",  
color="Sector")+  
theme_minimal()
```

Interpretation

- X-axis = Revenue
- Y-axis = Profit Margin
- Colour = Industry Sector
- Bubble Size = Market Capitalization
- Cartesian coordinates allow accurate comparison.
- Distinct colours reduce perceptual bias and separate sectors clearly.
- Bubble size effectively represents company scale through Market Capitalization.

